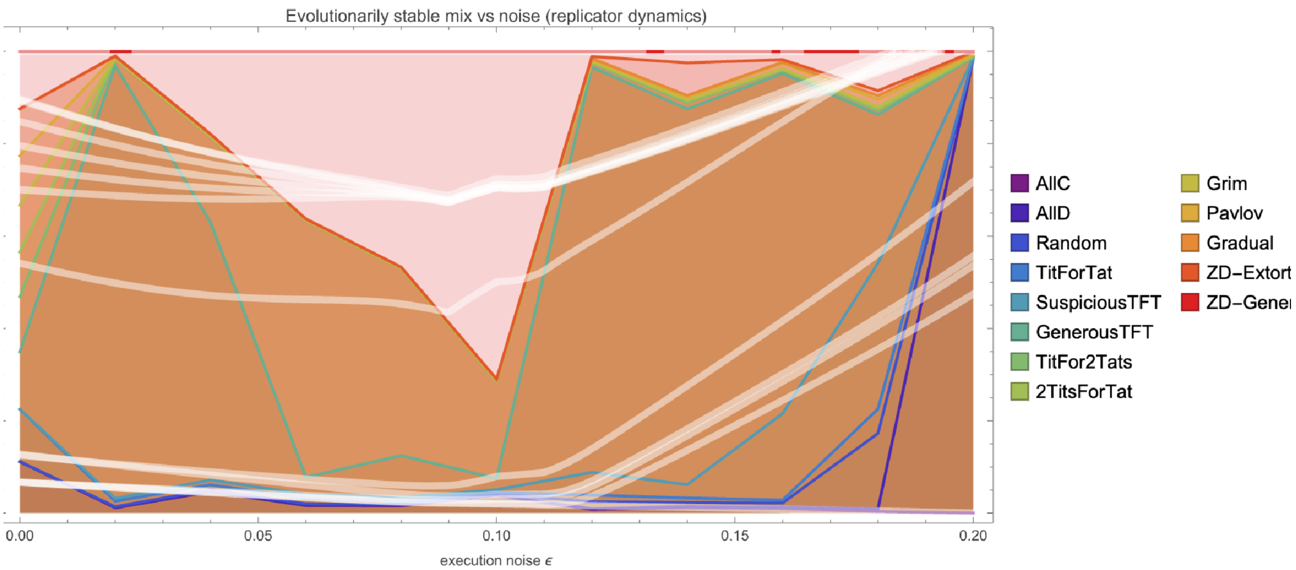
Re-running Axelrod's Tournament: Cooperation, Evolution, and Noise

The iterated Prisoner's Dilemma in the Wolfram Language – from Tit-for-Tat to an evolutionarily stable mix that depends on noise

Wolfram Community submission – May 2026



The headline result. As execution noise ϵ rises (left to right), the evolutionarily stable composition of strategies shifts – from a coexistence of generous reciprocators to a collapse into unconditional defection.

Abstract

In 1980 Robert Axelrod ran a now-famous computer tournament for the **iterated Prisoner's Dilemma**. The winner was the simplest entrant of all – **Tit-for-Tat**: cooperate first, then copy your opponent. This notebook reproduces that tournament in the Wolfram Language, with the classic strategies plus modern entrants (Generous Tit-for-Tat, Pavlov, Gradual, and Press – Dyson zero-determinant strategies).

It then does two things Axelrod's setup hints at but does not show. **First**, an *evolutionary* tournament: we let populations of strategies reproduce in proportion to their success and ask which mixture is stable. **Second**, we add *noise* – a small probability that an intended move is mis-executed – and show that the evolutionarily stable mixture **depends on the**

noise level. Finally, in the spirit of an automated-research loop, we let a search algorithm *discover* memory-one strategies that beat Tit-for-Tat under noise.

1. The Prisoner's Dilemma, and why iteration changes everything

Two players each choose, simultaneously, to **cooperate** (C) or **defect** (D). The payoffs satisfy $T > R > P > S$ and $2R > T + S$. With the classic Axelrod values $T=5, R=3, P=1, S=0$:

	opp C	opp D
I C	$R = 3$	$S = 0$
I D	$T = 5$	$P = 1$

In a *single* game, defection strictly dominates: whatever the opponent does, you score more by defecting. Yet mutual cooperation (R=3 each) beats mutual defection (P=1 each). That is the dilemma. Axelrod's insight was that when the game is **repeated**, the shadow of the future can sustain cooperation – a strategy can reward cooperation and punish defection in later rounds.

Encoding strategies in the Wolfram Language

We encode `C = 1` and `D = 0`. A strategy is a pure function of the two histories of actual moves; the smallest interesting family is the **memory-one** strategies, parameterised by the probability of cooperating after each of the four possible previous outcomes:

```
memOne[open_, {pCC_, pCD_, pDC_, pDD_}] := Function[{me, opp},
  If[me === {},
    If[RandomReal[] < open, 1, 0],
    If[RandomReal[] < {pCC, pCD, pDC, pDD}[[ stateIndex[Last[me], Last[opp]] ]], 1, 0]]];

(* Tit-for-Tat: cooperate first, then copy the opponent's last move *)
TitForTat = memOne[1, {1, 0, 1, 0}];
Pavlov    = memOne[1, {1, 0, 0, 1}];   (* win-stay, lose-shift *)
GenerousTFT = memOne[1, {1, 1/3, 1, 1/3}];
```

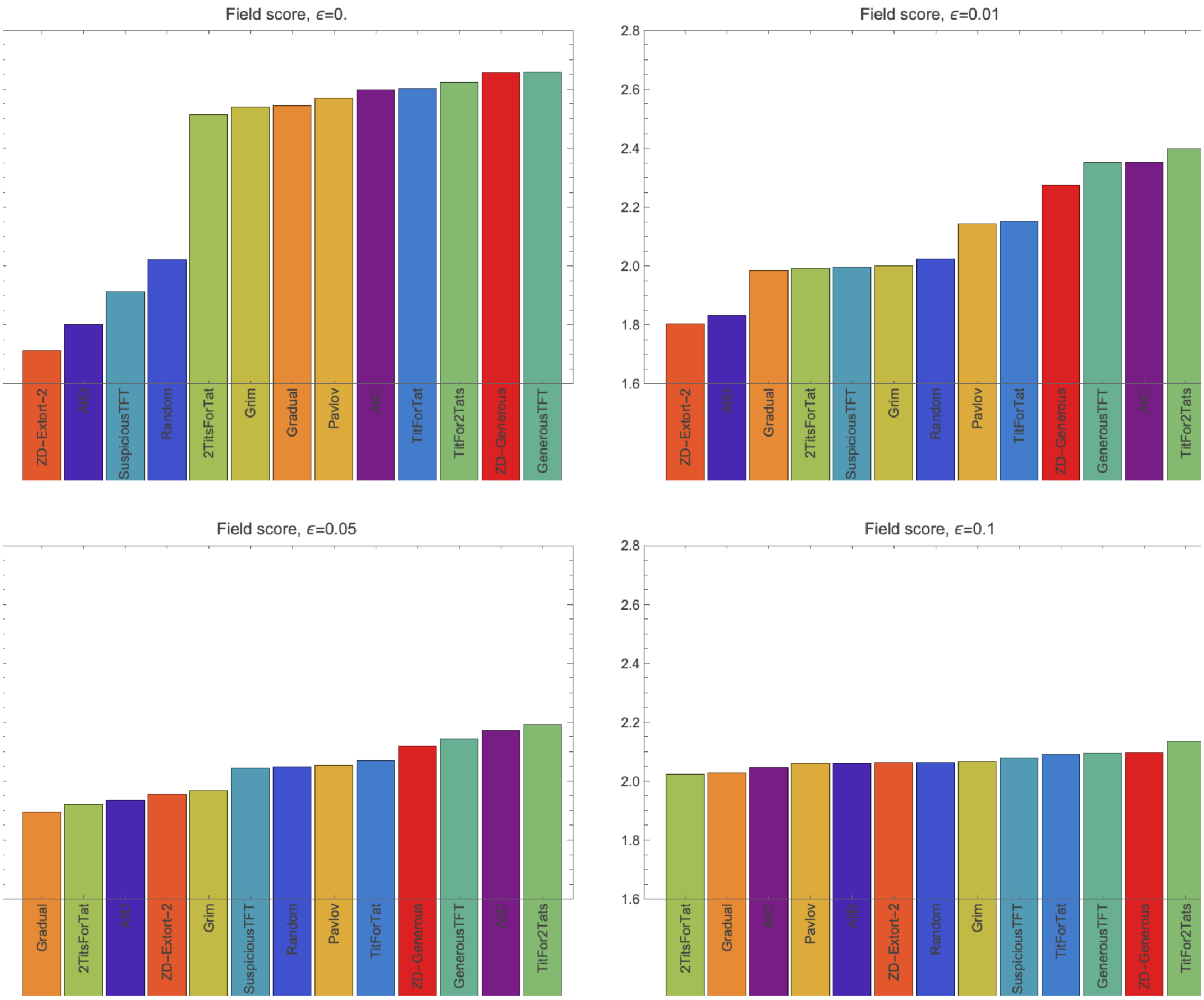
The full library (in `wolfram/axelrod.wl`) has 13 strategies: AllC, AllD, Random, Tit-for-Tat, Suspicious TFT, Generous TFT, Tit-for-2-Tats, 2-Tits-for-Tat, Grim, Pavlov, Gradual, and two zero-determinant strategies (an extortioner and a generous one).

Noise: the trembling hand

Real interactions are imperfect. We model **execution noise**: each intended move is flipped with probability ϵ, and the opponent sees the *actual* (post-noise) move. This single knob turns out to control everything that follows.

2. The classic round-robin (reproducing Axelrod 1980)

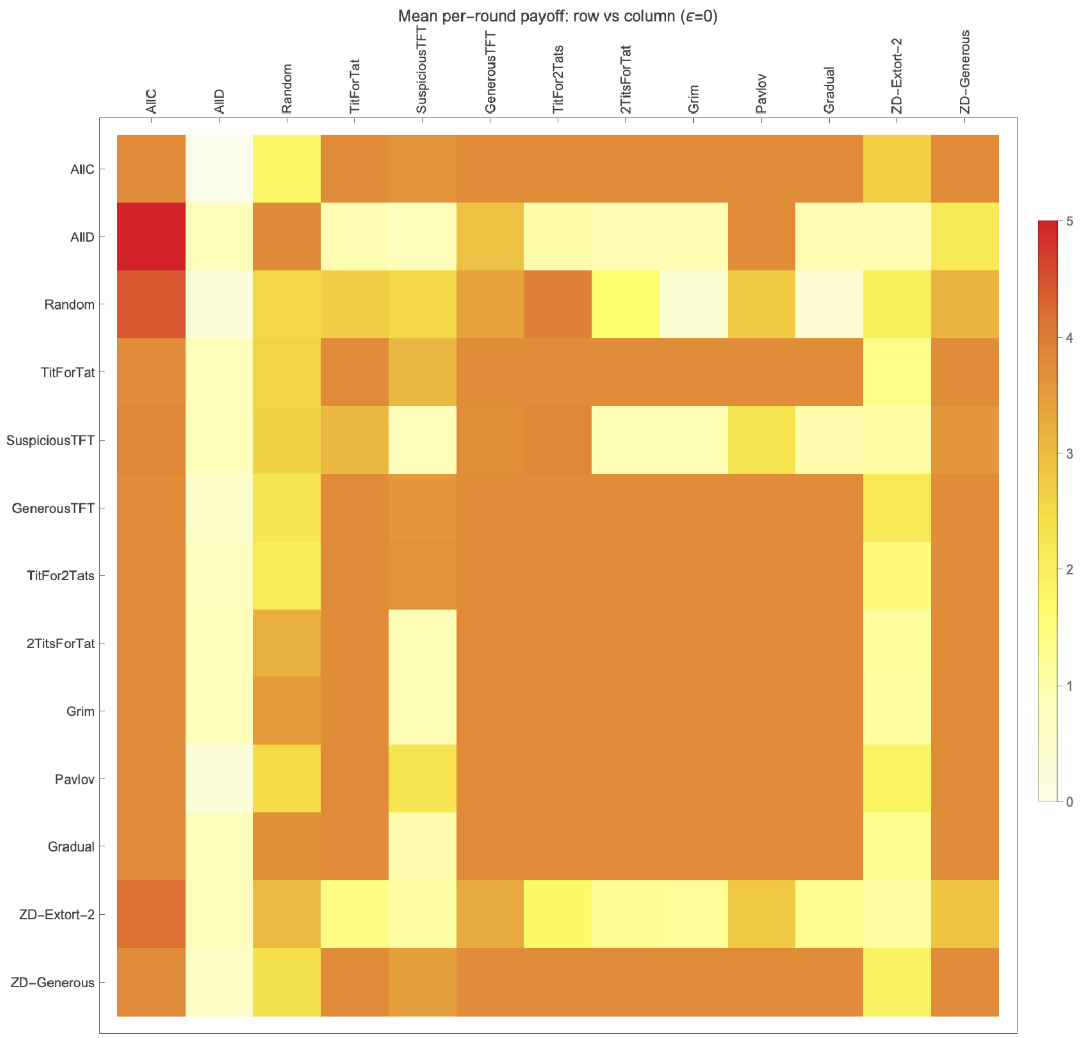
Every strategy plays every strategy (including itself) for 200 rounds, repeated 100 times to average out randomness. Each strategy's score is its mean per-round payoff against the whole field. At $\epsilon = 0$ (no noise) the ranking is:



Field score of each strategy at four noise levels. At ϵ=0 the generous reciprocators lead; the extortioner (ZD-Extort-2) finishes last.

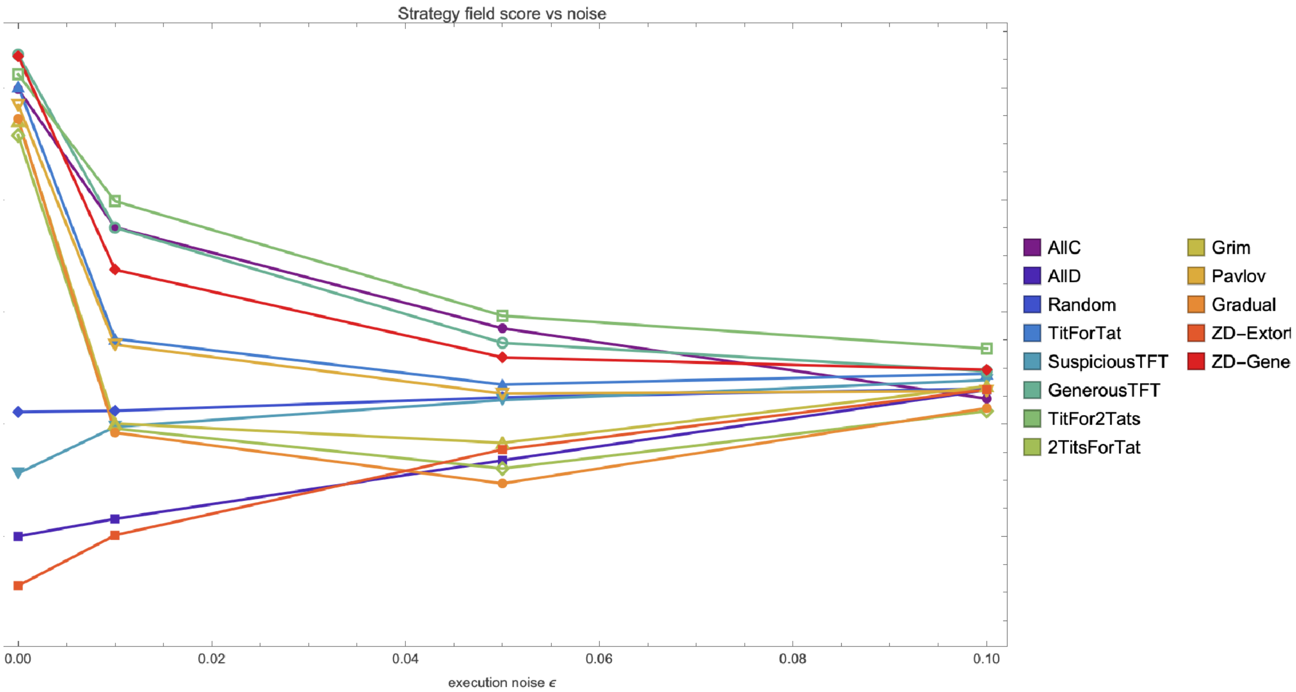
The classic lesson holds. Nice, reciprocal, forgiving strategies dominate. Notably, plain Tit-for-Tat is *edged out* by its more forgiving cousins – Generous TFT and Tit-for-2-Tats – exactly the refinement Axelrod found in his second tournament. And the zero-determinant **extortioner finishes dead last**: extortion wins individual matches but poisons the cooperative environment it depends on.

The payoff structure of who-beats-whom:



Mean per-round payoff, row strategy against column strategy (ϵ=0). Warm = high. The cooperative cluster (top-left block) sustains mutual R=3; AllD and the extortioner do well in single matchups but drag the whole field down.

What noise does to the ranking



As noise rises, every strategy's score is pulled toward the mutual-defection payoff P=1, and the gap between 'nice' and 'nasty' strategies narrows. Unforgiving strategies (Grim, Gradual) fall fastest: a single accidental defection traps them in retaliation.

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3. The evolutionary tournament – the stable mix depends on noise

Axelrod's deeper question was evolutionary: if successful strategies reproduce, which ones survive? We answer it two ways.

Replicator dynamics

Let x be the vector of population fractions and A the mean-payoff matrix ($A[i,j]$ = score of i against j). Each strategy grows in proportion to its fitness relative to the population average:

$$x_i' = x_i \left((A.x)_i - x.A.x\right)$$

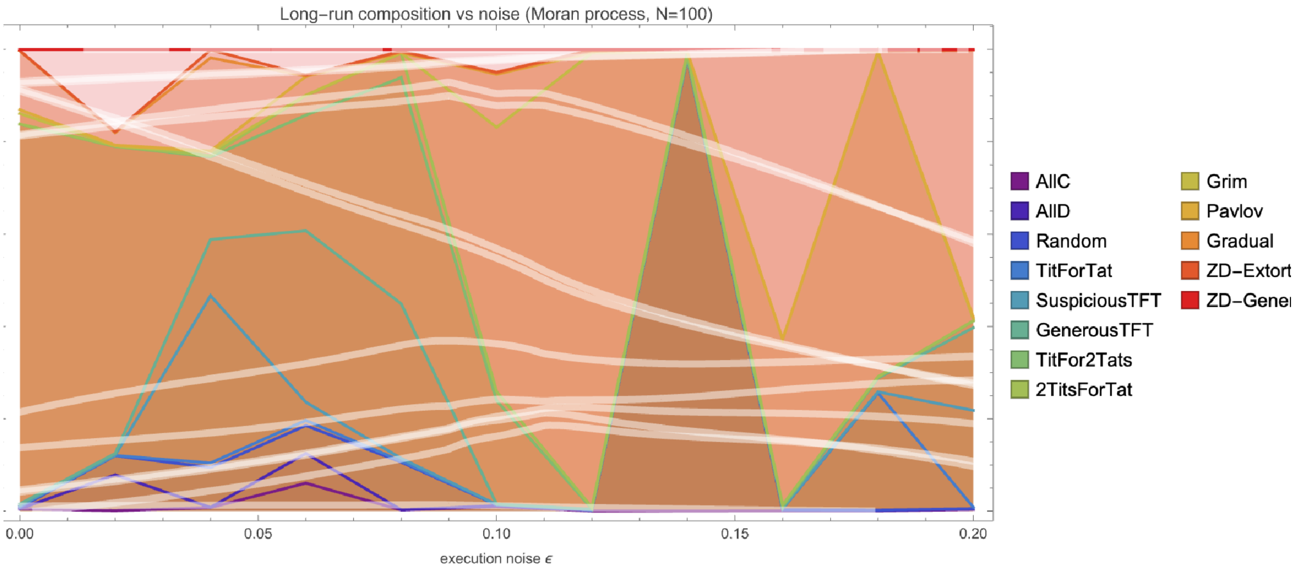
```
replicator[A_, steps_, mut_] := Module[{x = ConstantArray[1./n, n], f, phi},
  Do[ f = A . x; phi = x . f;
      x = x f / phi;              (* selection           *)
      x = (1 - mut) x + mut/n;    (* small mutation term  *)
      x = x / Total[x],
    {steps}];
  x]
```

Sweeping the noise level and recording the stable composition gives the headline figure (top of this post). It reveals **three regimes**:

- **Low noise ($\epsilon \approx 0$):** a coexistence of cooperative strategies – Generous TFT, Tit-for-2-Tats, ZD-Generous, Tit-for-Tat and Pavlov share the population. Cooperation is evolutionarily robust.
- **Intermediate noise:** the forgiving reciprocators (Generous TFT, Tit-for-2-Tats, ZD-Generous) dominate and trade the lead among themselves – error-correction is at a premium.
- **High noise ($\epsilon \geq 0.2$):** cooperation collapses. AllD sweeps to ~99% of the population: under heavy noise, reciprocators spend so long accidentally punishing each other that unconditional defection becomes the only stable attractor.

A finite-population check: the Moran process

Replicator dynamics is a deterministic, infinite-population limit. To confirm the result is not an artefact of that idealisation, we also run a stochastic **Moran process** (population N=100, frequency-dependent birth-death with rare mutation) and record the long-run average composition.



Moran process, long-run composition vs noise. Finite-population stochasticity makes the picture grainier, but the qualitative story survives: forgiving cooperators at low-to-moderate noise, defection-dominated outcomes as noise grows.

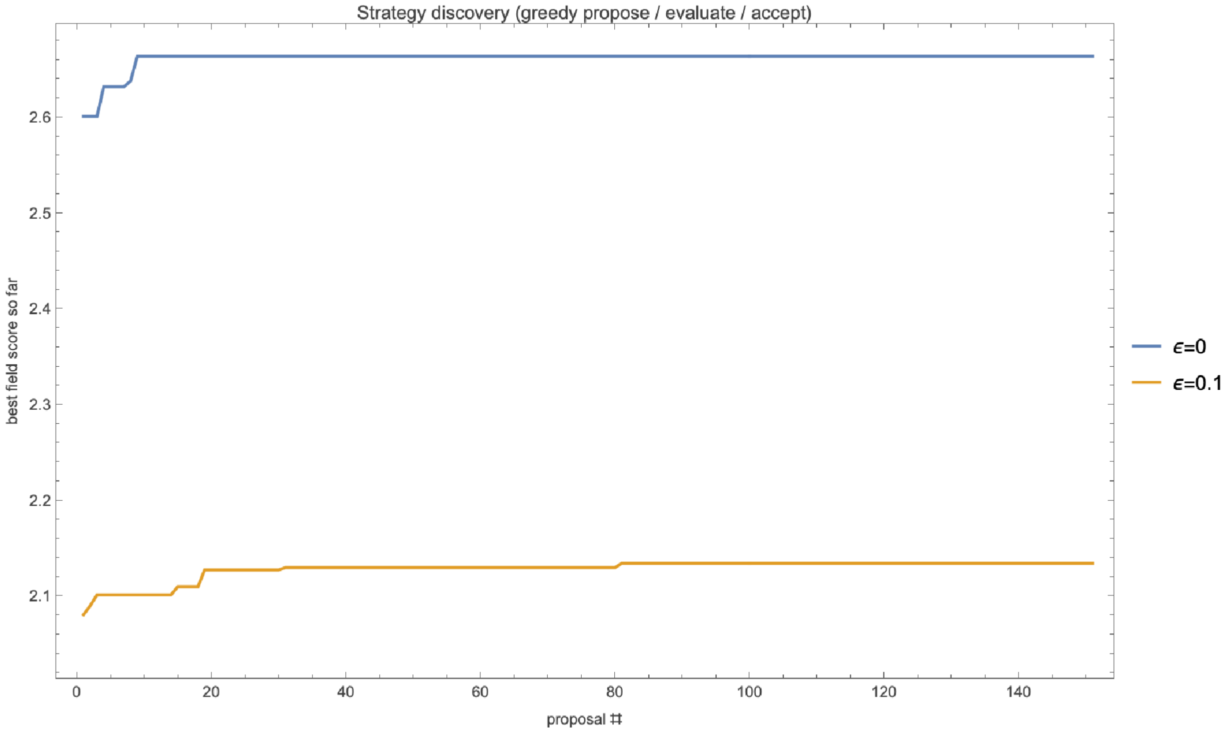
Takeaway. "Which strategy is evolutionarily stable?" has *no noise-independent answer*. The same population that converges on generous reciprocity in a clean channel converges on pure defection in a noisy one.

4. Discovering strategies that beat Tit-for-Tat under noise

Can we do better than the textbook strategies? In the spirit of an automated-research loop – *propose* → *evaluate* → *accept* → *repeat* – we search the memory-one space $\theta = (open, pCC, pCD, pDC, pDD) \in [0,1]^5$ for the best response to the standard field at each noise level. A greedy hill-climb gives the narrative; a small genetic algorithm is the workhorse.

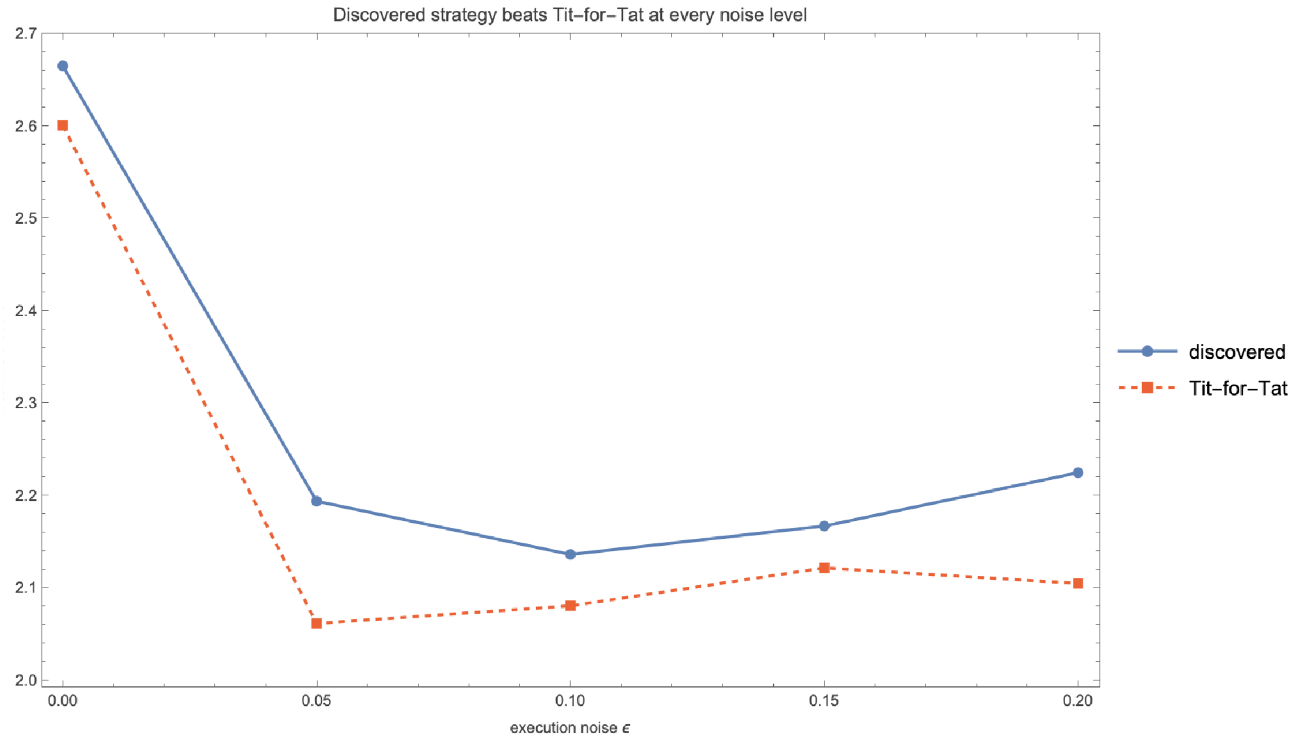
```
fitness[theta_, eps_] := Mean[
   meanMatch[memOne[theta[[1]], theta[[2;;]]], #, rounds, eps, reps][[1]] & /@ field];

(* greedy accept/reject loop -- the autoresearch analogy *)
champion = {1, 1, 0, 1, 0};                       (* start from Tit-for-Tat *)
Do[ proposal = clip[champion + RandomVariate[NormalDistribution[0, .12], 5]];
    If[fitness[proposal, eps] > fitness[champion, eps], champion = proposal],
  {150}]
```



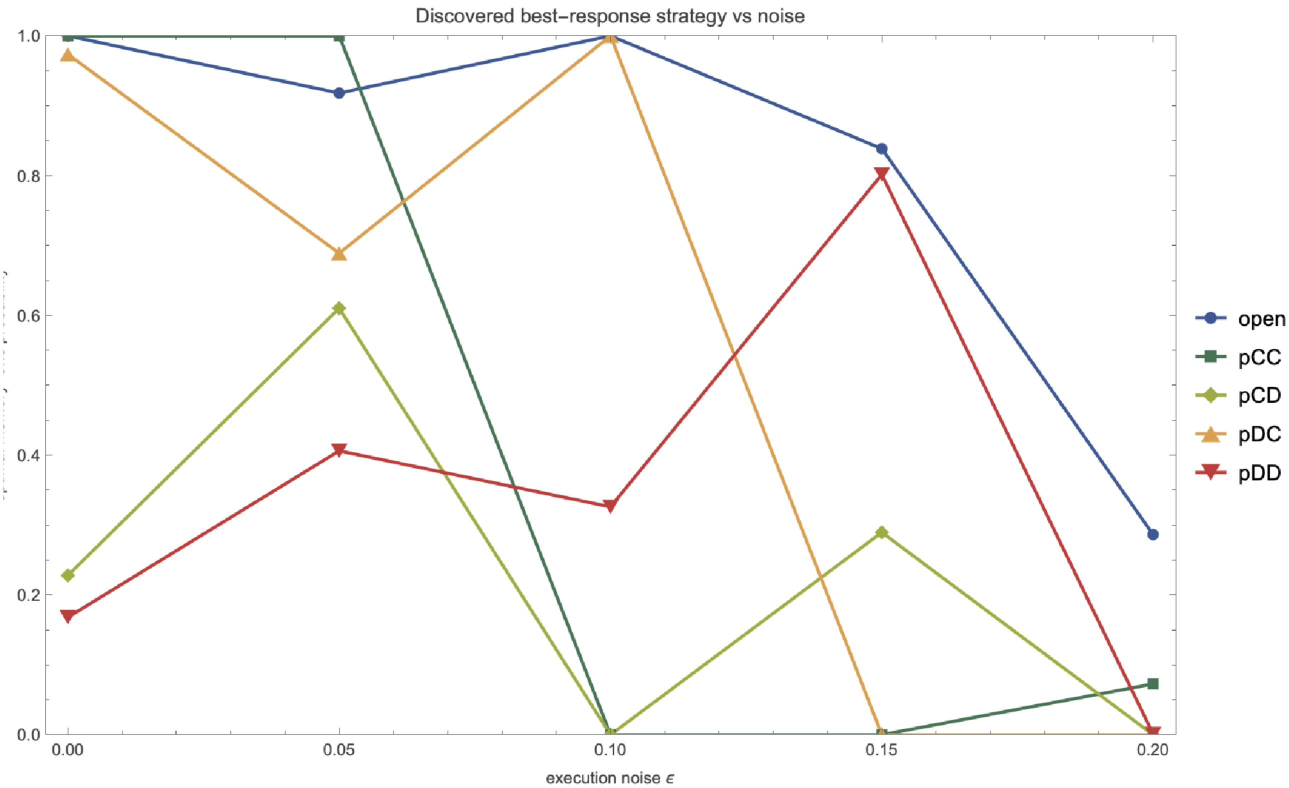
Greedy discovery: best field score found so far, starting from Tit-for-Tat, at two noise levels. Each upward step is an accepted proposal.

The discovered strategies beat Tit-for-Tat at every noise level



Discovered best-response strategy (solid) vs plain Tit-for-Tat (dashed) in field score, across noise. The discovered strategy wins everywhere; the margin is largest where noise matters most.

And the *shape* of the optimal strategy drifts with noise:



The discovered memory-one probabilities vs noise. The forgiveness parameter pCD (cooperate again after being suckered) rises with noise – error-correction becomes essential – until, at very high noise, the best response against a decaying field tilts back toward defection.

This is the same lesson as the evolutionary tournament, reached by a different route: **the optimal behaviour is a function of the environment**, and noise is the knob that turns generous reciprocity into defection.

5. Letting an automated-research loop invent strategies

The discovery in §4 searched a continuous parameter box. We can also run the loop the way a human researcher would: **propose a named hypothesis, evaluate it, keep it if it wins, repeat** – exactly the shape of Karpathy-style *autoresearch* scaffolds (an AI edits one solution file; a fixed harness scores it; improvements are committed). We ported that loop to this domain: `autoresearch/evaluate.wls` is the fixed harness (an 18-strategy field × six noise levels) and a sequence of hypotheses is proposed, each scored by mean field score averaged over the noise sweep.

```
champion <- Tit-for-Tat
LOOP:  propose a strategy  ->  evaluate vs the field under noise
       if score > champion: keep (save to champions/)  else: revert
       append (iter, score, status, description) to results.tsv
```

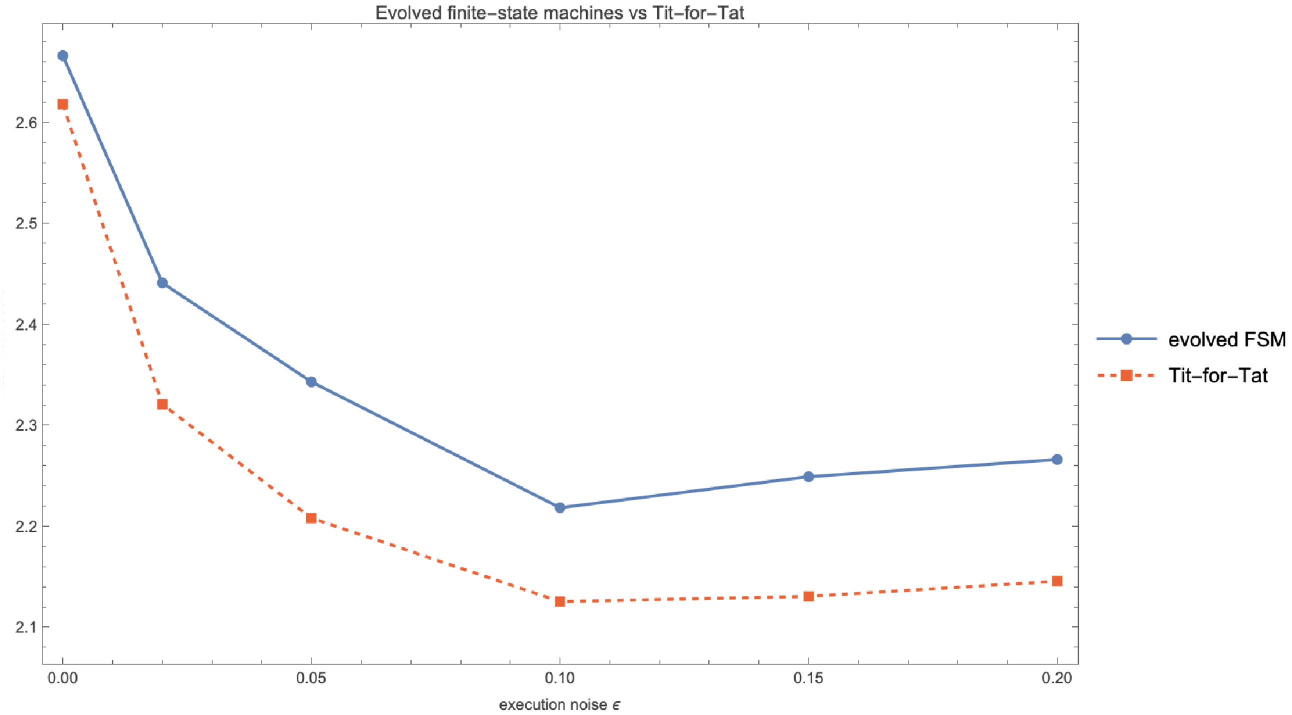
The result is a research log. Against this field (TFT baseline `2.316`), the loop found **seven strategies that beat Tit-for-Tat**. The winners are instructive:

- **Tit-for-2-Tats (2.400).** Patience beats vigilance: absorbing a single (possibly accidental) defection before retaliating is the single biggest win.
- **Generous Tit-for-2-Tats and Generous TFT.** Tolerance plus probabilistic forgiveness compounds the gain.
- **AdaptiveGenerous – a self-tuning entrant.** It estimates the channel noise online (from how often the opponent's moves flip) and scales its own forgiveness to match – beating TFT with no hand-set forgiveness constant.

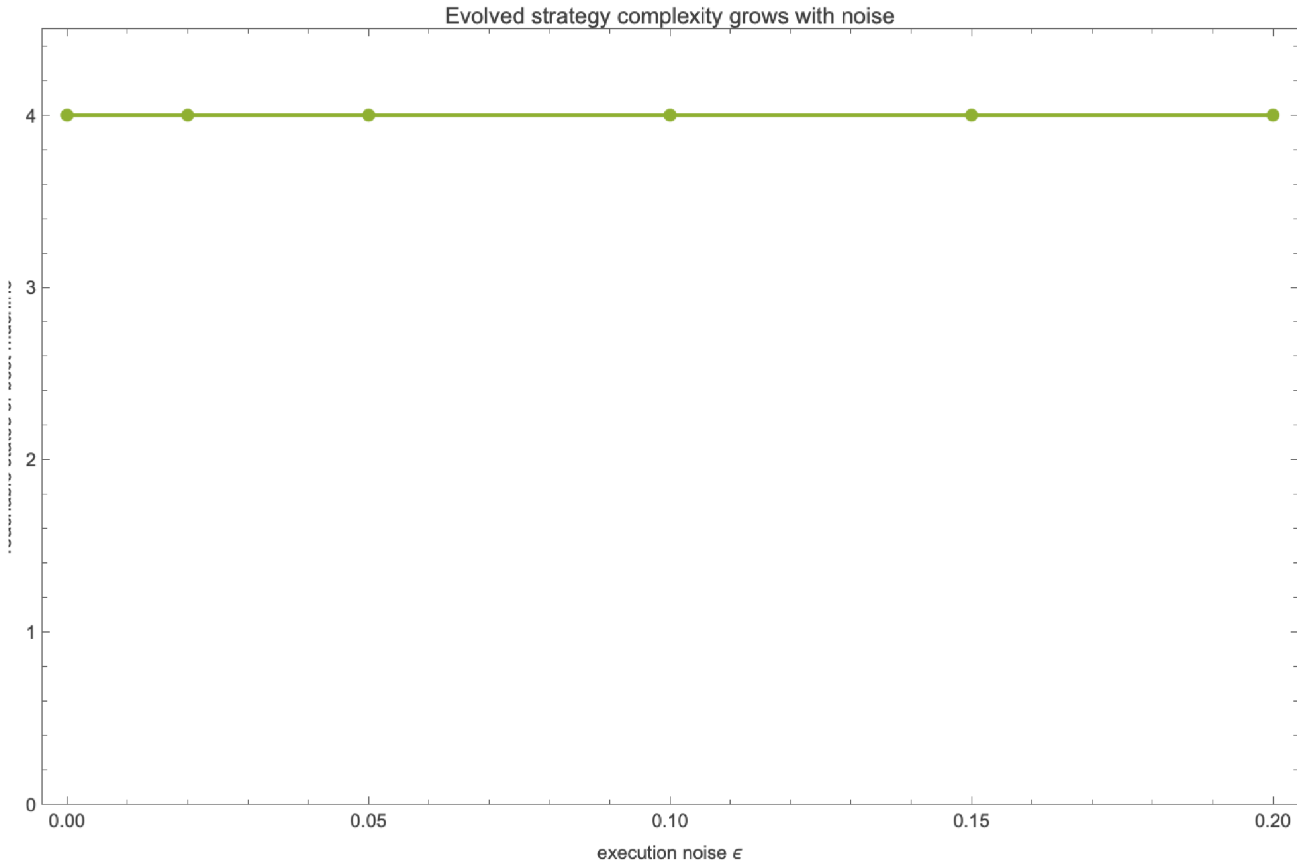
What lost is as telling as what won. Plain contrition and Win-Stay-Lose-Shift (Pavlov) scored *below* TFT in this well-mixed field – forgiveness has to be *patient* (two-strikes), not merely reflexive. Hold that thought: Pavlov's fortunes change once we add spatial structure.

6. Beyond memory-one: evolving finite-state machines

Memory-one strategies condition only on the previous round. Do richer strategies help? We let a genetic algorithm evolve **finite-state (Mealy) machines** – each a little automaton with up to four internal states, an action and a state-transition for every observed opponent move. This space contains TFT, Grim, Pavlov and much else; we verified it encodes TFT and Grim exactly before evolving it.



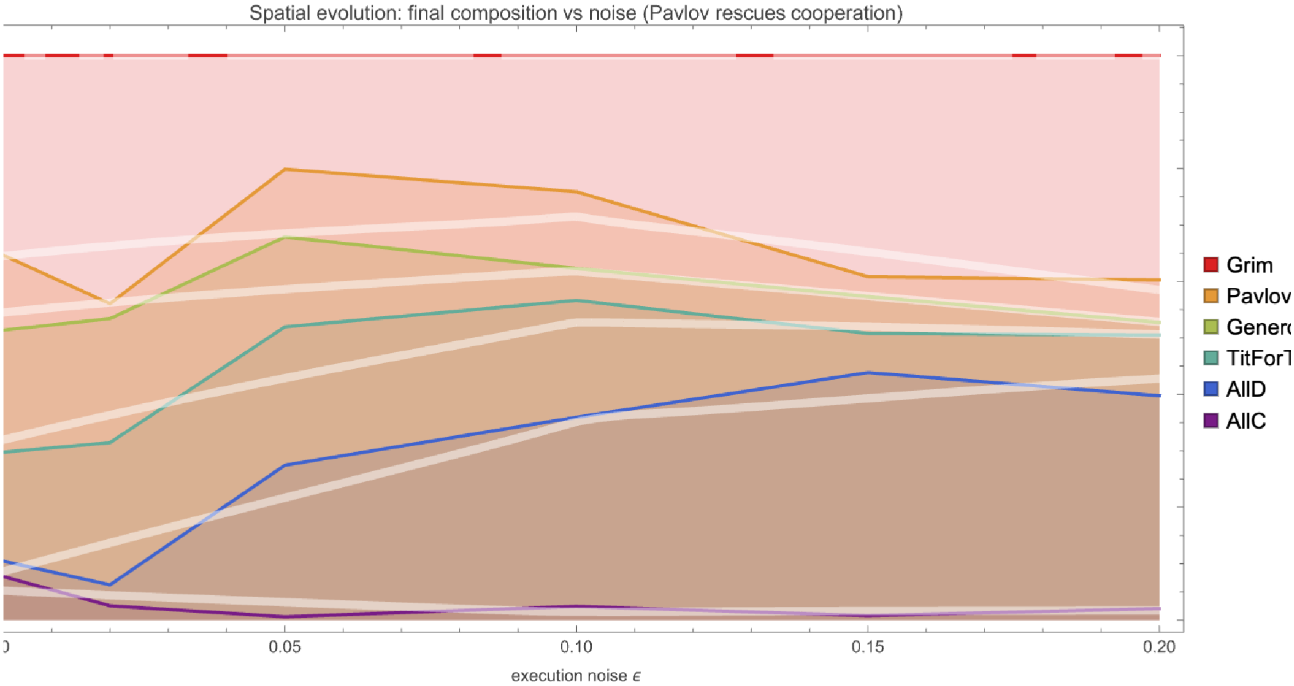
Evolved finite-state machines vs Tit-for-Tat across noise – the machines beat TFT at essentially every noise level.



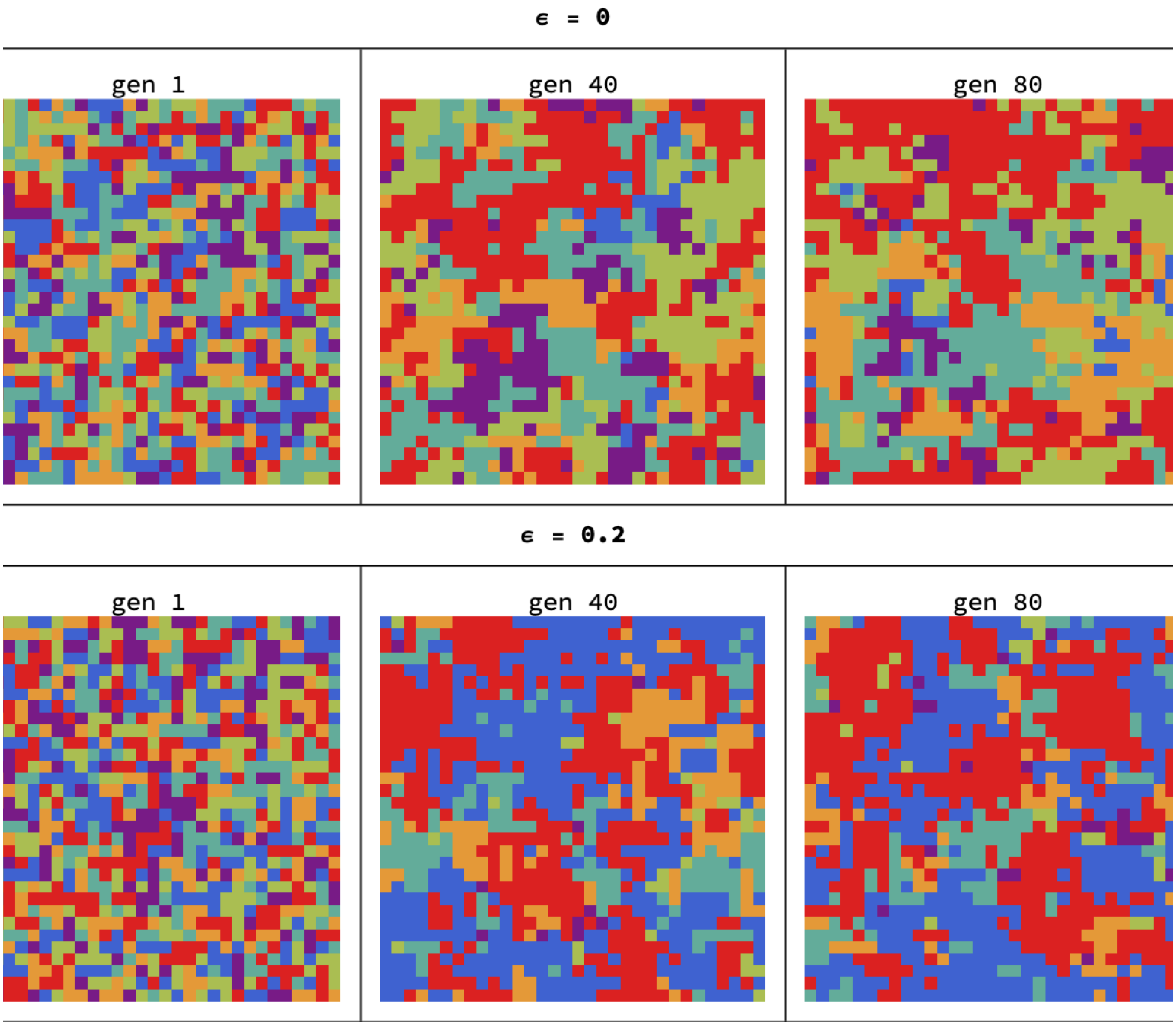
The best evolved machine uses 2–4 of its 4 reachable states – genuine multi-state memory, beyond what memory-one can express. The gain over the memory-one optimum is real but modest: against this field, most of the advantage is already captured by one round of memory plus the right amount of forgiveness.

7. Spatial structure makes cooperation robust to noise

Everything so far assumed a *well-mixed* population. But real agents interact **locally**. We place strategies on the sites of a 32×32 lattice (a torus). Each generation, every site plays its four neighbours and imitates fitter neighbours (Fermi rule). This is Nowak – May spatial game theory; the classic result is that clustering lets cooperators survive. We ask what noise does to it.



Final lattice composition vs noise. At low noise the lattice is almost entirely cooperative reciprocators (Grim, Generous TFT, Tit-for-Tat), with unconditional defection nearly absent. As noise rises AllD gains ground – but never takes over: cooperation persists around 60% even at ϵ=0.2.

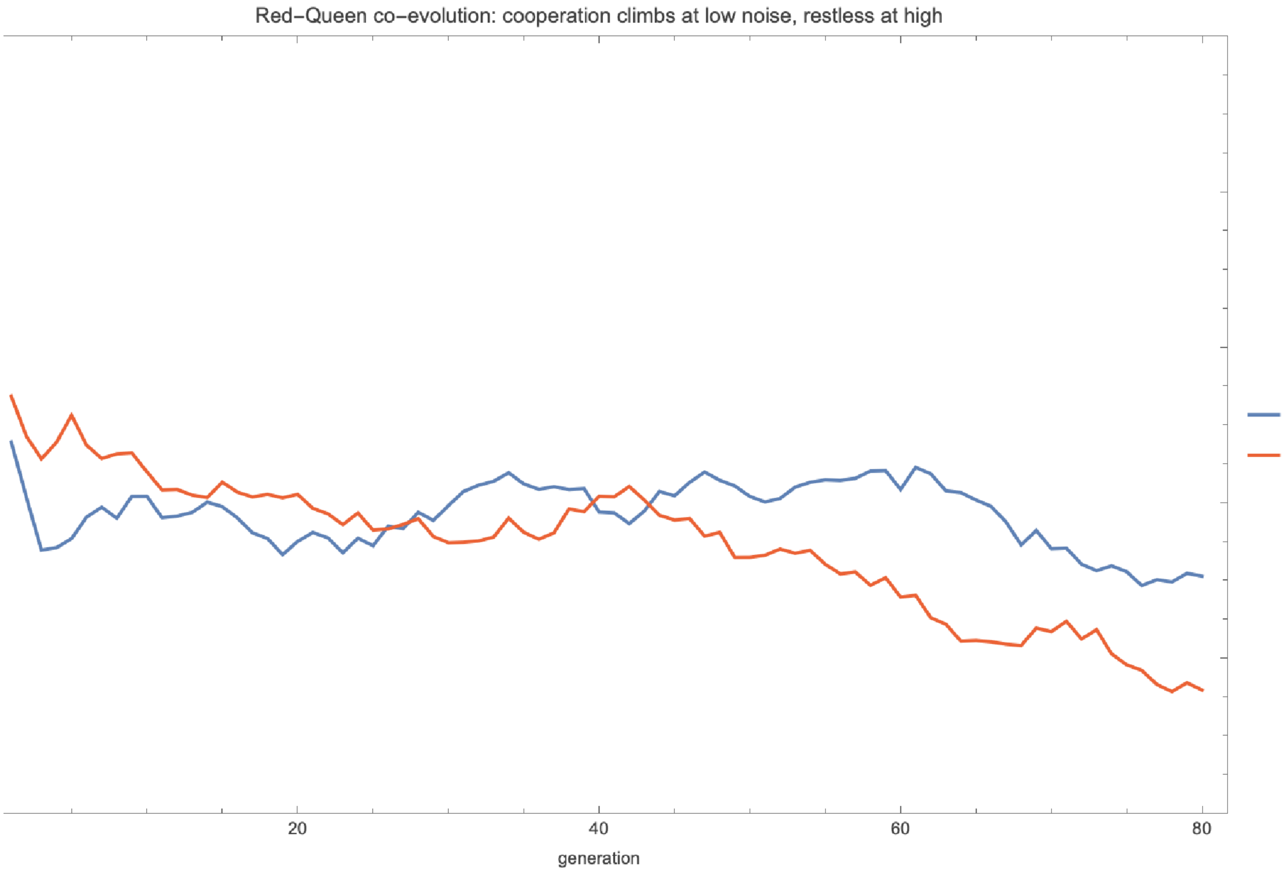


Lattice snapshots (colours = strategies; legend below). Reciprocators survive by forming contiguous clusters that protect their cooperative interior. Noise frays the cluster boundaries but does not dissolve them.

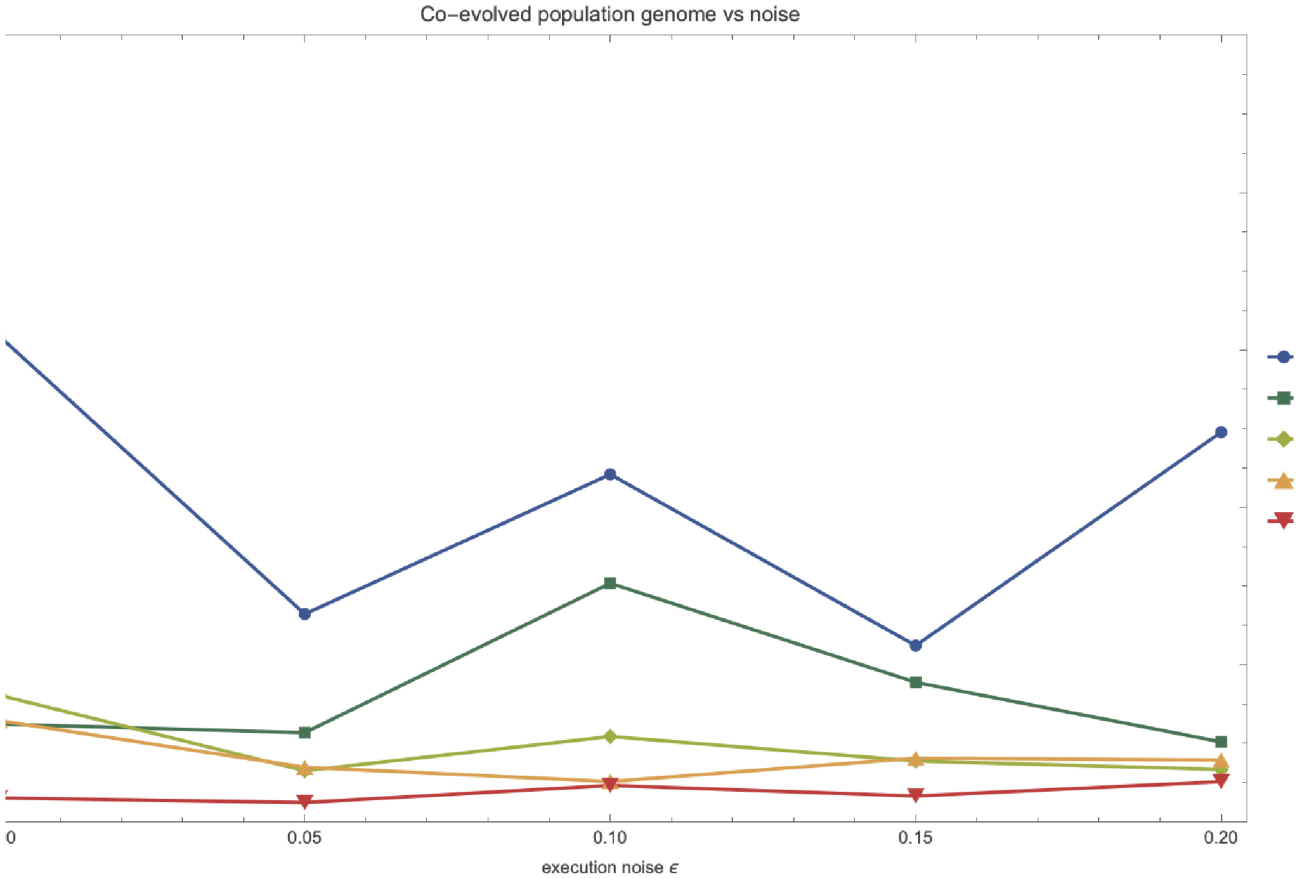
Spatial structure makes cooperation far more robust to noise. In the well-mixed model (§3), once ϵ is large the population collapses almost entirely to defection. On the lattice the same noise erodes cooperation only partially: clustering lets reciprocators shield one another, so cooperation degrades gracefully (97% → ~60%) instead of collapsing. **Whether noise is catastrophic or merely corrosive for cooperation depends on population structure.**

8. Red-Queen co-evolution

So far strategies were optimised against a *fixed* field. What if the field itself evolves? We co-evolve a population of memory-one genomes against **itself**: each individual's fitness is its score against the current population, which changes every generation. This is a *Red Queen* race – you must keep adapting just to hold your rank.



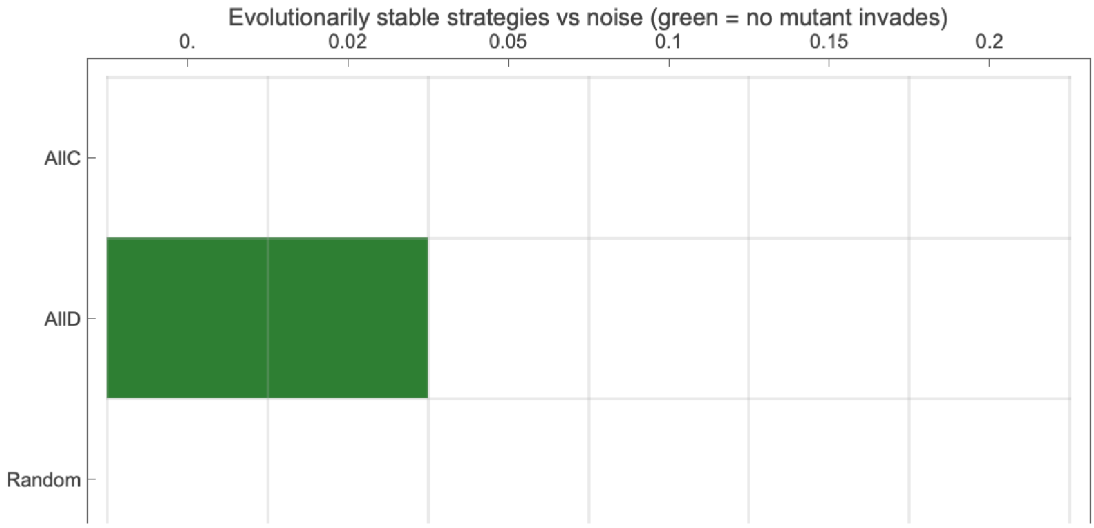
Population-mean cooperativeness over generations. At ϵ=0 the population climbs to stable, highly cooperative reciprocity. At ϵ=0.15 it rises, then becomes restless and slides back – noise prevents a stable cooperative equilibrium from locking in.

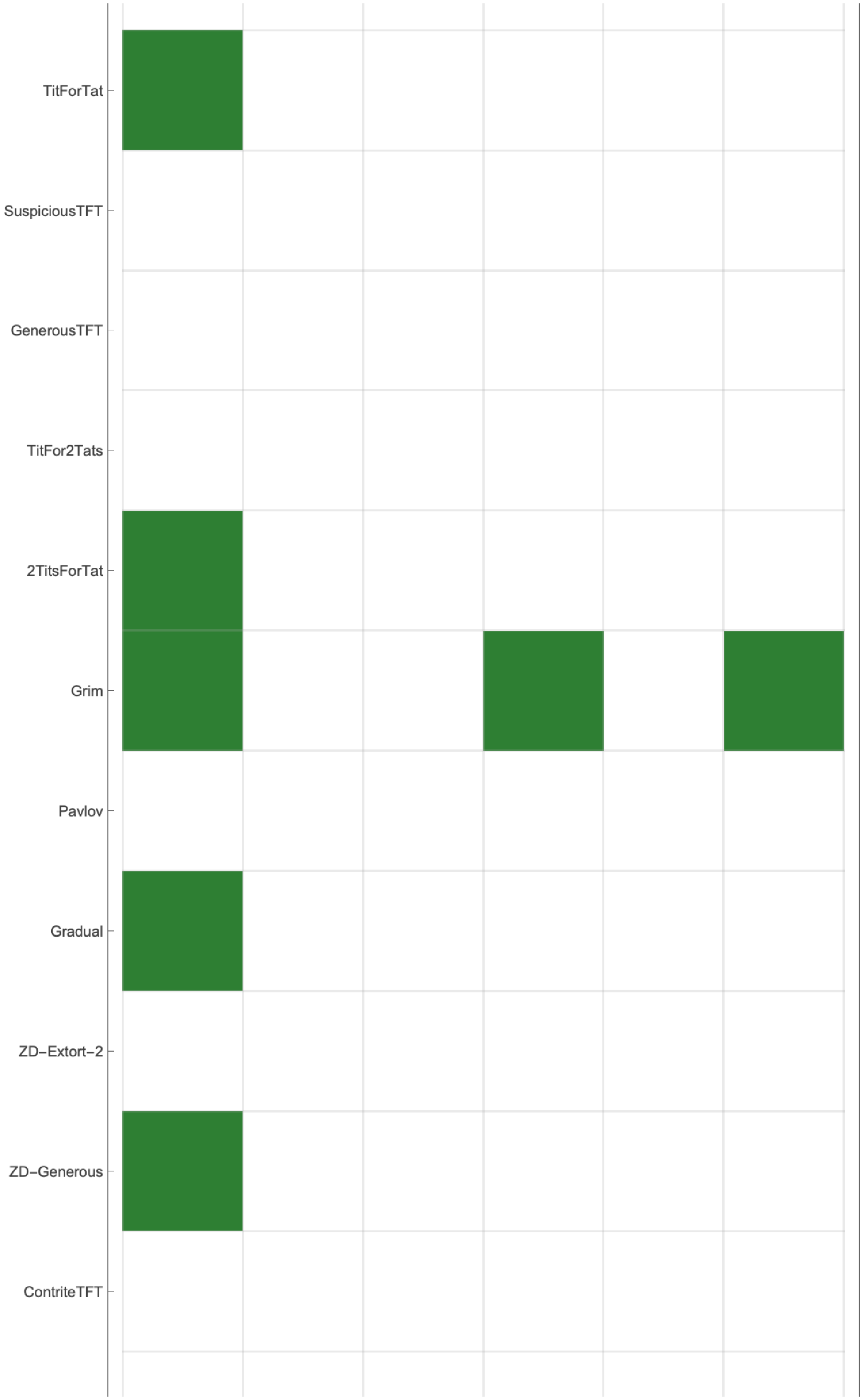


The co-evolved mean genome vs noise. As noise rises the population becomes less rigidly cooperative (pCC falls) but more forgiving (pCD rises) and more able to recover from mutual defection (pDD rises) – a third independent route to the same conclusion: forgiveness answers noise.

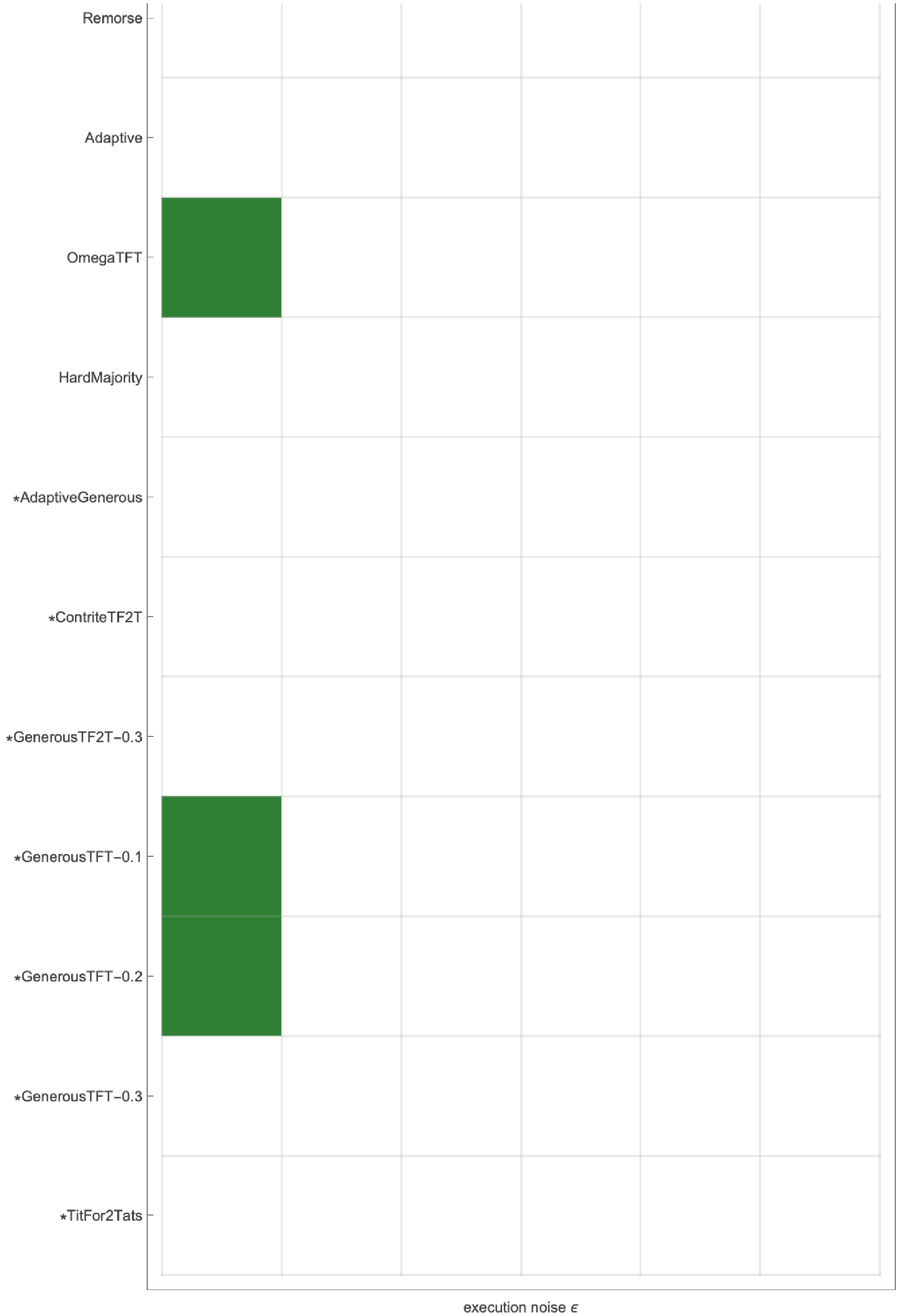
9. Which strategies are evolutionarily stable?

A rigorous capstone. For a panel of 25 strategies (classics, modern entrants, and the discovered champions) we compute, at each noise level, the pairwise **invasion matrix**: can a rare mutant M invade a resident population of R? It can if $A[M,R] > A[R,R]$. A strategy is **evolutionarily stable** if no mutant can invade it.



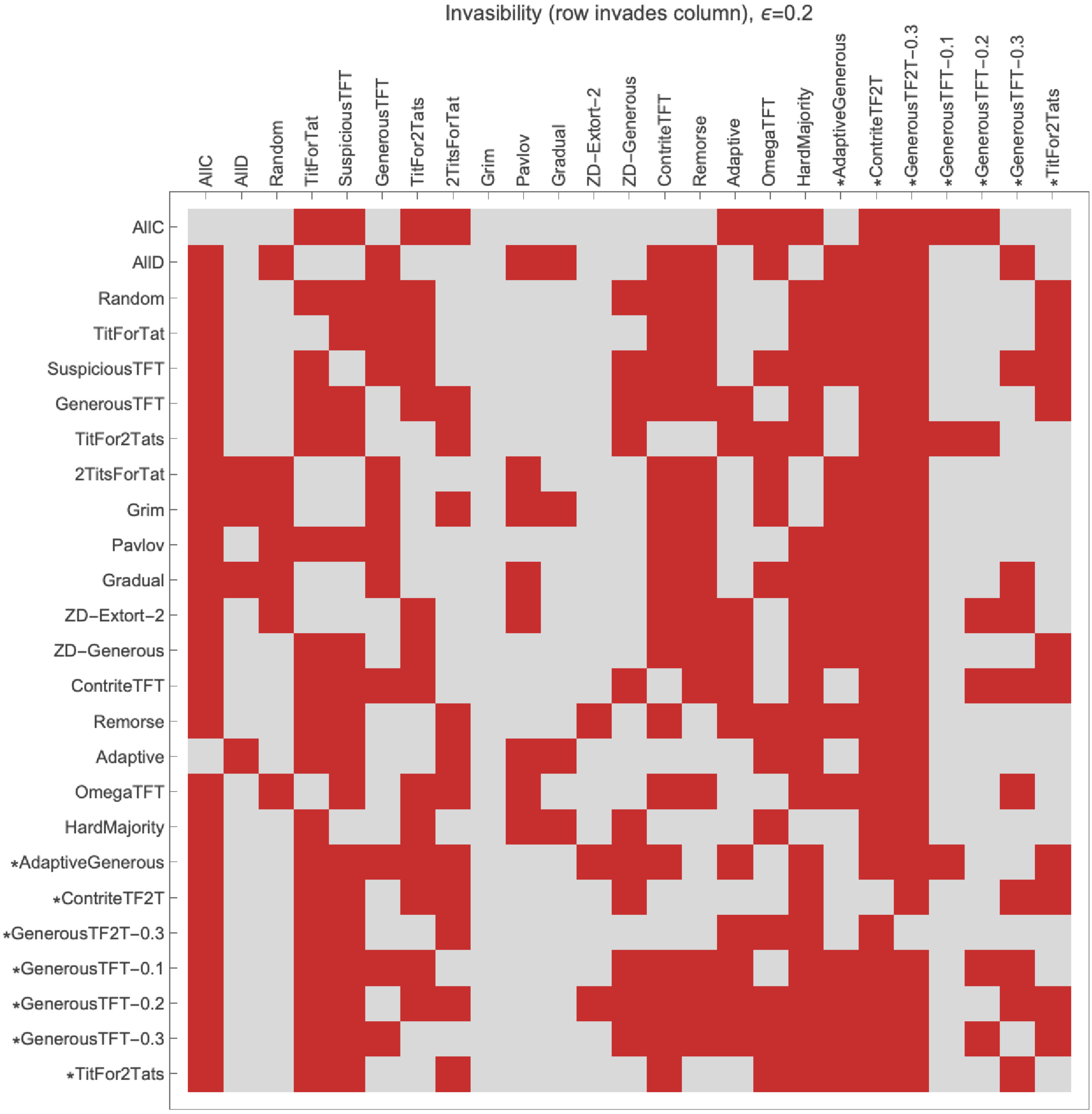


TitForTat
SuspiciousTFT
GenerousTFT
TitFor2Tats
2TitsForTat
Grim
Pavlov
Gradual
ZD−Extort−2
ZD−Generous
ContriteTFT




```
captionCell["Evolutionarily stable strategies vs noise (green = no mutant invades). Only ",
 StyleBox["defectors", FontWeight → Bold],
```

```
" are strict ESS: at ϵ=0 both AllD and the extortionate zero-determinant strategy resist
  invasion; the instant noise appears, AllD alone survives the test. No forgiving
  or cooperative strategy is an ESS at any noise level – cooperation is never
  safe from invasion by a lone defector, however well it does in a population."]
```



Invasibility matrix at ϵ=0.2 (red = row invades column). The defectors have the cleanest columns.

Three meanings of "stable." "Which strategy is stable?" has *three* answers: (i) **ESS** (uninvadable) favours the defectors, more so under noise; (ii) **replicator-stable from a cooperative start** (§3) gives generous reciprocity collapsing under noise; (iii) **spatially resilient** (§7) gives reciprocator clusters that survive noise. All three are noise-dependent and they disagree – there is no single best strategy, only best-for-a-given-world.

10. Conclusions

- **Tit-for-Tat is excellent but not optimal.** In a cooperative field, more forgiving reciprocators (Generous TFT, Tit-for-2-Tats) edge it out – and extortion finishes last.
- **Forgiveness must be patient.** The autoresearch loop found two-strikes tolerance (Tit-for-2-Tats) beats every reflexive scheme; a self-tuning strategy that estimates the noise also beats TFT.
- **“Stable” has three answers.** ESS favours defectors; replicator-from-cooperative-start gives generous reciprocity collapsing under noise; spatial structure gives reciprocator clusters that survive. All noise-dependent, and they disagree.
- **Population structure changes the effect of noise.** Well-mixed: noise drives cooperation to collapse. Spatial: clustering keeps it resilient, degrading gracefully instead of collapsing.
- **Many roads, one lesson.** Fixed-target search, finite-state GP, replicator and Moran dynamics, open-ended co-evolution, and ESS analysis all converge: forgiveness is the answer to noise – until noise is so high that nothing sustains cooperation in a well-mixed world.

Reproducing this work

Everything here is a few seconds to a few minutes of Wolfram Language – no external data, no API keys. The full source is on GitHub.

```
wolframscript -file tests/sanity.wls            # correctness checks
wolframscript -file wolfram/tournament.wls      # round-robin ranking
wolframscript -file wolfram/evolution.wls       # replicator + Moran vs noise
wolframscript -file wolfram/discover.wls         # memory-one discovery
wolframscript -file autoresearch/run_search.wls  # autoresearch champions
wolframscript -file wolfram/fsm_discover.wls     # evolve finite-state machines
wolframscript -file wolfram/spatial.wls          # spatial / lattice evolution
wolframscript -file wolfram/coevolution.wls      # Red-Queen co-evolution
wolframscript -file wolfram/ess.wls              # ESS invasion analysis
wolframscript -file wolfram/figures.wls          # regenerate base figures
```

Built entirely in the Wolfram Language. The strategy-discovery and autoresearch loops borrow the propose → evaluate → accept shape of automated-research scaffolds, re-implemented natively in WL.